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Influence of Music Education and Pitch Scales on the Grouping of the AB-AB Sequence Sounds

Introduction

Sound perception is related to two types of processes in the human auditory system – sensory and cognitive processes. Sensory processes refer to the set of phenomena associated with the reception of acoustic waves from the environment, the processing of acoustic waves into nerve impulses in the auditory organ, the encoding in the form of impulses of data about the physical characteristics of the received sounds and the transmission of nerve impulses to the hearing centres in the cerebral cortex. The effectiveness of sensory processes related to hearing is determined by the physiological state of the human auditory system. Research in the field of psychoacoustics, which examines the relationship between the physical characteristics of sound and auditory sensation, usually assumes that there are no major differences in the mechanisms of these phenomena between people with physiologically normal hearing.

Cognitive processes, on the contrary, are related to the processing by the mind of information coming from the organ of hearing and other senses. Processing occurs in the central nervous system and involves receiving information from the environment, storing and transforming it, and reintroducing it into the environment in the form of a response-behaviour. Cognitive processes include, among others, attention, awareness, perception, memory, thinking and reasoning[1]. The resulting cognitive structures are used by the individual to form a mental image of the stimuli received.

1 E. Nęcka, J. Orzechowski, B. Szymura, *Psychologia poznawcza*, Academica Wydawnictwo SWPS, PWN, Warszawa 2006, pp. 178, 278, 320, 420.

Cognitive processes depend on various factors related to the past experiences of the person receiving sensory stimuli. For this reason, people with a similar physiological state of hearing but with different experiences of sound perception, may differ in, for example, their sensitivity to sound changes and perceive auditory images with different properties and structure when their hearing organ is stimulated by the same acoustic signal. Experiences acquired as a result of musical education and gained in professional music practice, among others, have a significant impact on the formation of cognitive processes in sound perception.

As a result of the hearing process, a variety of perceptual elements and structures can arise in the mind of the hearer. When listening to a sound, it is possible to identify sound sources, the acoustic properties of the environment in which the sources are located and to differentiate sounds in terms of their elementary sensory attributes, which include loudness, pitch, timbre and perceived duration. Traditional research into the principles of sound perception uses specialised methods of auditory evaluation to obtain a description of the relationship between the physical characteristics of sound and the qualities of the impression received by the hearer.

Research studies in psychoacoustics and music acoustics employ the concept of a perceptual stream. A perceptual stream (or an auditory or acoustic stream) is a sequence or a set of auditory impressions arranged in such a manner that the hearer perceives a given acoustic event as an entity. This means that the recipient can, for instance, recognise and distinguish individual sounds as voices in polyphonic works or harmonic structures (distinguishing different perceptual streams) through their ability to find elements with similar characteristics, coherence or regularity. The essence of the perceptual stream is to extract distinctive characteristics of sounds from their entirety in order to obtain an overall picture or a complete auditory description of a given acoustic event.

The literature provides evidence that the cognitive processes involved in sound perception are developed in musicians in a specific manner, which makes musicians different from non-musicians in sound perception situations not specifically related to music. Based on the results of published research, it can be concluded that musical education affects sound perception in situations where cognitive processes play a predominant role in sound perception. A phenomenon in which cognitive processes play a particularly important role is combining sounds into perceptual streams.

The research presented in this study was conducted in order to determine whether the processes of grouping sounds into perceptual streams demonstrate different characteristics in musicians and non-musicians. The nature of the research is experimental and its essential part is the psychoacoustic experiment conducted by the author.

Experimental assumptions and aim

Experiment aimed to investigate what influence on perceptual stream grouping is exerted by the pitch differences of sound pairs played in sequence in different pitch ranges. The experiment was based on those described by Miller and Heise[2], Dowling[3], Bregman and Campbell[4], Bregman[5], and van Noorden[6]. It investigated the phenomenon of sound grouping into perceptual streams based on the pitch difference between two sounds, depending on the tempo of sequence playback[7]. The sound pairs were played in different octaves. In addition, previous pilot experiments conducted by this author[8] with two cohorts of participants, i.e. musicians and non-musicians, demonstrate that the two groups display certain noticeable and simultaneously evident differences in the perception of the same sound stimuli, which inspired further and much more in-depth research concerned with the analysis of auditory imaging in the context of perceptual streaming.

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2 G.A. Miller, G.A. Heise, *The Trill Threshold*, "Journal of Acoustical Society of America" 1950, vol. 22, issue 1, pp. 637, 638.

3 W.J. Dowling, *Rhythmic Fission and Perceptual Organization*, "Journal of Acoustical Society of America" 1968, vol. 44, issue 1, p. 369.

4 A.S. Bregman, J. Campbell, *Primary Auditory Stream Segregation and Perception of Order in Rapid Sequences of Tones*, "Journal of Experimental Psychology" 1971, vol. 89, no. 2, pp. 244–249.

5 A.S. Bregman, *Auditory Scene Analysis. The Perceptual Organization of Sound*, The MIT Press, Cambridge, Massachusetts 1990, pp. 65–67, 157–163.

6 L.P.A.S. van Noorden, *Minimum Differences of Level and Frequency for Perceptual Fission of Tone Sequences ABAB*, "Journal of Acoustical Society of America" 1977, vol. 61, no. 4, pp. 1041–1045; idem, *Temporal Coherence in the Perception of Tones Sequences* (unpublished dissertation), Technical University Eindhoven, Eindhoven 1975, pp. 1–4, 7–10, 18–20, 28–32.

7 A.S. Bregman, *Audio Demonstrations of Auditory Scene Analysis*, http://webpages.mcgill.ca/staff/Group2/abregm1/web/downloadstoc.htm [accessed: on 12.12.2022].

8 A. Rosiński, *Wpływ wykształcenia muzycznego na grupowanie dźwięków sekwencji ABA–ABA w rytm galopujący*, in: *Przestrzenie akustyki. Professional Acoustics*, A. Rosiński (ed.), Wydawnictwo Uniwersytetu Warmińsko-Mazurskiego w Olsztynie, Olsztyn 2021, pp. 53–70.

Sound material

The experiment used repeated sequences comprised of two sounds separated by a perfect fifth. This interval was played in three octaves: great, one-line, and three-line. Sound frequencies were set according to equal-temperament tuning:

– in the great octave: D = 73.416 Hz, A = 110.000 Hz,

– in the one-line octave: d^1 = 293.666 Hz, a^1 = 440.000 Hz,

– in the three-line octave: d^3 = 1174.700 Hz, a^3 = 1760.000 Hz.

The time plot of stimulus replay is depicted in Figure 1. Sounds were played in sequence: low sound and high sound in each series. The time between the beginning of one sound and the beginning of the neighbouring sound was taken as the basic measurement time step. A single time series in this experiment were two sounds played alternately, without intermissions, similar to the experiment conducted by Shonle and Horan[9].

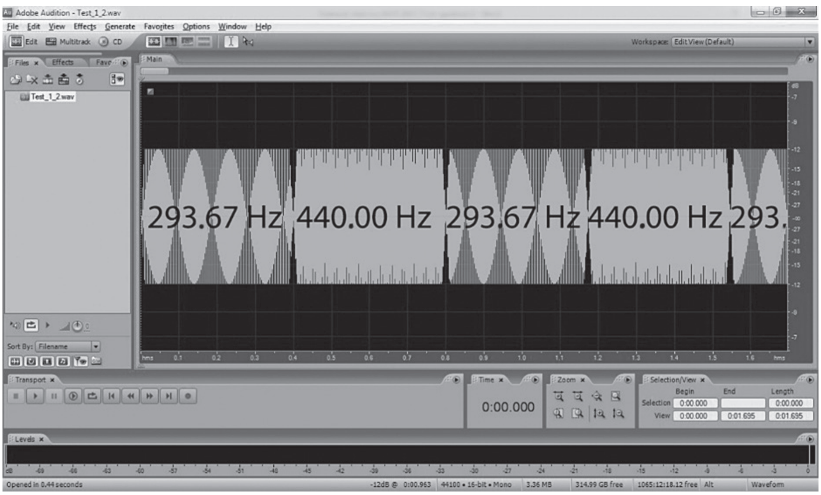

Fig. 1. Screen capture – an analysis by *Adobe Audition* software depicting a time paradigm of a sound sequence played in the experiment. Frequencies shown in the interface window relate to the interval played in one-line octave
Source: own research

[9] J.I. Shonle, K.E. Horan, *Trill Threshold Revisited*, "Journal of Acoustical Society of America" 1976, vol. 59, issue 2, pp. 469–471.

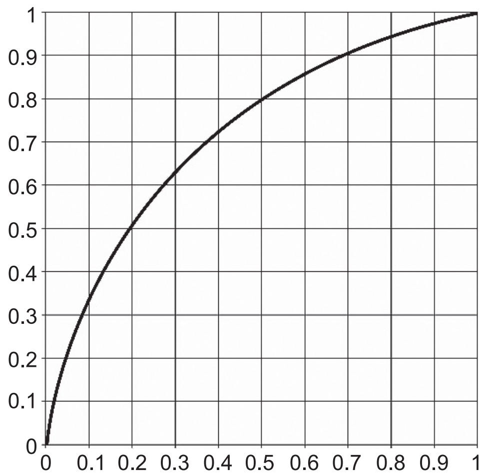

Fig. 2. Duration of sounds in AB sequences as a function of time elapsed from task start
Source: own study, software-generated analysis from: *Matemaks.pl. Matematyka maksymalnie prosta*. *Program do rysowania wykresów funkcji*, https://www.matemaks.pl/program-do-rysowania-wykresow-funkcji.html [accessed on: 9.02.2021]

The duration of each sound was computer-generated and lasted from 400 ms to 40 ms[10]. In order to avoid crackling in audio replay, the amplitude envelope of each sound was given a linear attack and release of 10 ms[11]. The temporal sequence of sounds was regulated and analysed by software on the WPF.NET platform.

Time control of sound duration was modelled after the sound material used in the experiment described by Bregman and Campbell[12]. The duration

10 A.S. Bregman, *Auditory Scene Analysis. The Perceptual Organization of Sound*, The MIT Press, Cambridge, Massachusetts 1990, pp. 157–158; ibidem, p. 50; L.P.A.S. van Noorden, op. cit., p. 1041; G.L. Dannenbring, A. S. Bregman, *Effect of Silence Between Tones on Auditory Stream Segregation*, "Journal of Acoustical Society of America" 1976, vol. 59, no. 4, p. 987.

11 G.L. Dannenbring, A.S. Bregman, *Stream Segregation and the Illusion of Overlap*, "Journal of Experimental Psychology. Human Perception and Performance" 1976, vol. 2, no. 4, pp. 546, 568.

12 A.S. Bregman, J. Campbell, *Primary Auditory Stream Segregation and Perception of Order in Rapid Sequences of Tones*, "Journal of Experimental Psychology" 1971, vol. 89, no. 2, pp. 244–249; A.S. Bregman, *Audio Demonstrations of Auditory Scene Analysis,* http://webpages.mcgill.ca/staff/Group2/abregm1/web/downloadstoc.htm [accessed: on 12.12.2022].

of sounds gradually decreased over the course of the task, following a hyperbolic function (illustrated in Fig. 2), according to the formula:

$$t = ((-1/(3x+1))+1) * (4/3)$$

where t signifies the duration of a single sample, and x signifies the time elapsed since the sequence start.

The series of sounds was presented a maximum of 207 times, which gives 414 sounds replayed in a single presentation. The task was conducted three times.

Hardware and software employed in listening sessions

Listening sessions were held in computer room 120 of the Gdańsk Academy of Fine Arts, and in chamber music hall S2 of the Gdańsk Academy of Music. The listening station comprised the following elements:

– *Asus M51VA-AP117* portable computer with *Windows 7 Ultimate 64-bit* operating system, and the *Microsoft* programming platform *Windows Presentation Foundation* (WPF.NET). The application was developed in the *Microsoft Visual Studio 2013* programming environment.
– *Beyerdynamic DT 770 pro* closed headset with 80-ohm impedance. The volume of the stimuli was determined in a previous pilot study involving three musicians and three non-musicians with no prior experience with auditory experiments. These individuals established a level of audio path amplification that supplied the volume of maximum comfort during the listening experiment. The established loudness level was approximately 70 phons. All audio samples were generated on the WPF.NET software platform with the *Microsoft Visual Studio 2013* development environment, using a mono system with a 44.1 kHz sampling rate and a resolution of 16 bits. All analyses and statistical calculations were carried out using *IBM SPSS Statistics V23.0 software.*

Subjects

The experiment involved 48 subjects, aged 21 to 27, including 24 students or graduates of the Gdańsk Academy of Music and 24 students or graduates of the Gdańsk Academy of Fine Arts. The group of 48 subjects was comprised of 40 women and 8 men. Candidates for the groups were recruited at random.

The study was designed as a series of experimental sessions, all of which involved two groups of subjects: students and young graduates of the Academy of Music, and students and young graduates of the Academy of Fine Arts who were non-musicians. For the purposes of this paper, students of the Gdańsk Academy of Music and young professional musicians are jointly referred to as "musicians", while "non-musicians" are the students of the Academy of Fine Arts, including individuals with some experience in amateur music. Detailed data of participants' education, professional speciality and music experience are presented below.

Data concerning musicians

1. Subjects began their education between the ages of 5 and 10, and graduated from primary and secondary music school.
2. They continuously practiced playing musical instruments over the last 13–22 years.
3. Professional education of three subjects was temporarily interrupted for incidental reasons.
4. The subjects did not possess absolute pitch.
5. The subjects mostly performed classical music.
6. All subjects were right-handed.

Data concerning non-musicians

1. Five subjects had attended a music centre, primary music school, or private lessons in singing or instrumental music. Those individuals received a few years of music education, and some of them were involved in amateur music performance.
2. Other subjects received no instrumental or singing instruction, nor performed music as amateurs.
3. The majority of non-musicians listened to general popular music.
4. All were right-handed.

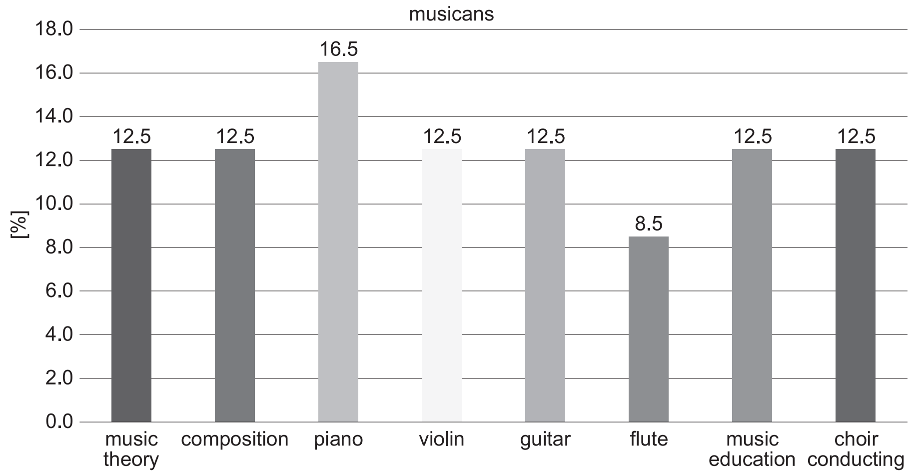

Fig. 3 . Percentage of subjects among musicians by academic focus or professional specialty
Source: own research

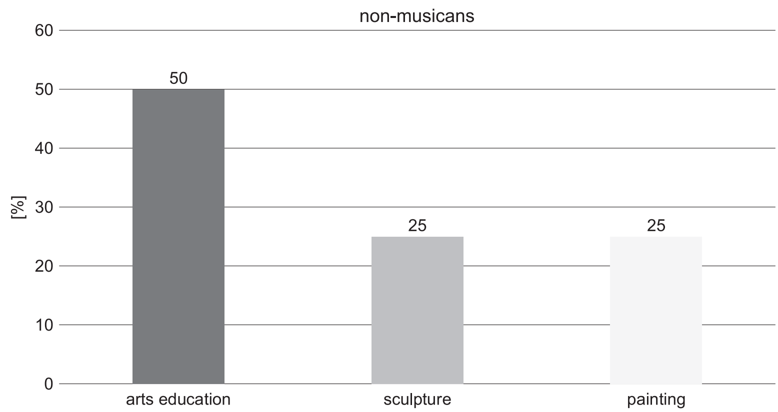

Fig. 4. Percentages of non-musicians by academic focus or professional specialty
Source: own research

Listening sessions

The study was conducted in listening sessions separate for each participant. The subjects were instructed to indicate the moment that they heard the sounds separate in two perceptual streams. At the beginning of the session, a demonstration series was played, taken from the work by Bregman[13],

[13] Ibidem.

in order to familiarise the listener with the phenomenon of perceptual fission and to explain the experimental task.

Results

Table 1 presents group results of the experiment – mean sample duration at each group's indication of perceptual stream fission, as well as standard deviation and standard error of each value set. The values shown in Table 1 are displayed in graphical form in Figure 5.

Table 1. Group results of experiment – mean sample durations (in ms) at which each listener group indicated sounds separating into two perceptual streams, as well as standard deviation and standard error of each value set. Results are divided by octave: great, one-line and three-line

Specification	Group	*N*	Mean	Standard deviation	Mean standard error
Great octave	musicians	24	114.25	27.355	5.584
	non-musicians	24	169.13	45.431	9.273
One-line octave	musicians	24	94.83	27.747	5.664
	non-musicians	24	179.29	49.849	10.175
Three-line octave	musicians	24	79.21	28.152	5.746
	non-musicians	24	172.00	56.083	11.448

Source: own research

Figure 5 presents responses given by musicians and non-musicians as to the duration of sample sounds (threshold) at which they experience fission of perceptual sound streams. The graph shows mean values and standard deviations of the sets of 24 responses in the musician group and non-musician group, divided by octaves: great, one-line, and three-line.

To investigate whether differences in the mean results between the groups of musicians and non-musicians carry statistical significance, a Student's t-test for independent samples was performed. The calculated t values and the associated p likelihood values are shown in Table 2. p values, shown in the rightmost column, are ($p < 0.001$), which is grounds for rejecting the null hypothesis that the average results in the two groups are not statistically different.

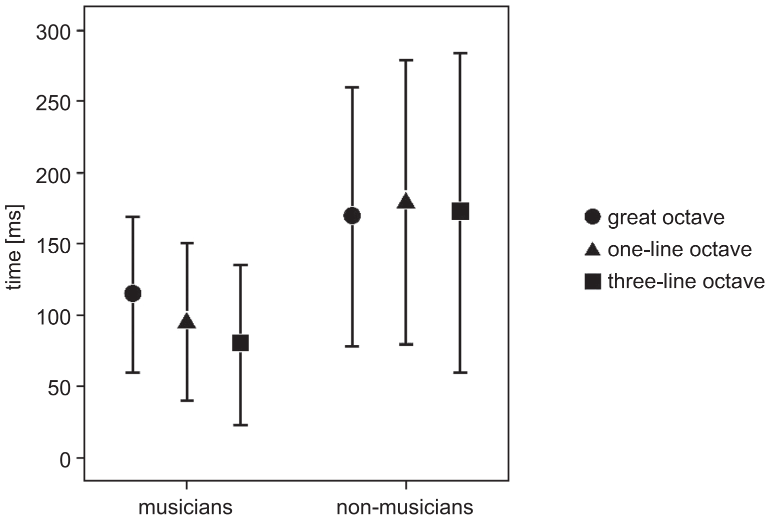

Fig. 5. Duration of samples [ms], at which sounds separated by a perfect fifth fissioned in two perceptual streams. Mean values and standard deviations of results by octave: great, one-line, three-line
Source: own research

Table 2. Results of Student's *t*-test for independent variables – experiment

Specification	*t*-test for equality of means		
	t	*df*	significance (two-tailed)
Great octave	-5.069	23	$p < 0.00001$
One-line octave	-7.252	23	$p < 0.000$
Three-line octave	-7.244	23	$p < 0.000$

Source: own research

Figures 6–8 show histograms of subject responses in tests regarding auditory stream fission in two perceptual streams. The set of 24 responses in the group of musicians and non-musicians was segregated in discrete time intervals. The *x*-axis shows the number of subjects whose responses fell in a particular interval. Subsequent figures show graphs relating to sequences played in great, one-line and three-line octaves, respectively.

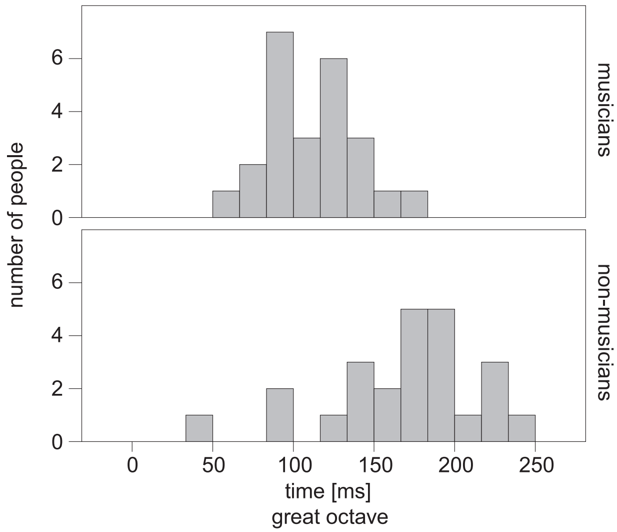

Fig. 6. Number of subjects – musicians and non-musicians whose responses fell in particular time intervals plotted on the *x*-axis.
Results for sound sequences played in the great octave
Source: own research

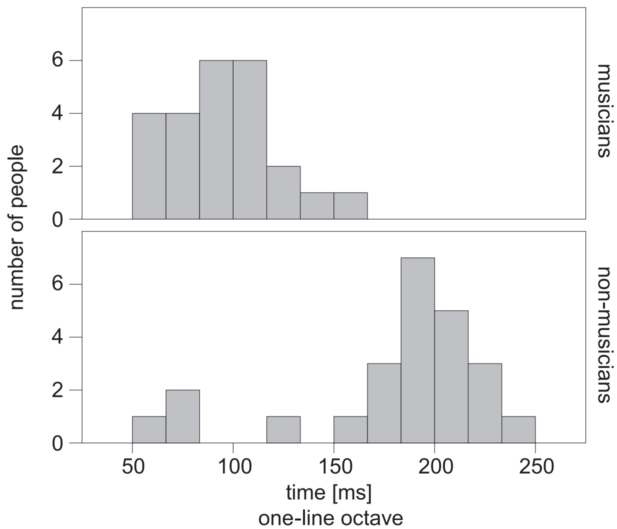

Fig. 7. Number of subjects – musicians and non-musicians whose responses fell in particular time intervals plotted on the *x*-axis.
Results for sound sequences played in the one-line octave
Source: own research

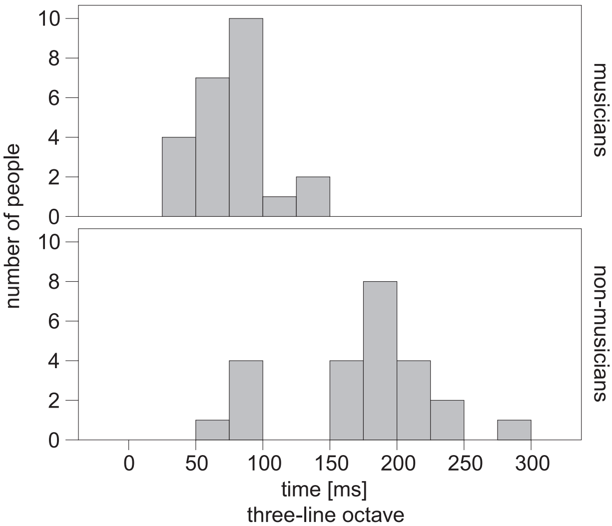

Fig. 8. Number of subjects – musicians and non-musicians whose responses fell in particular time intervals plotted on the *x*-axis.
Results for sound sequences played in the three-line octave
Source: own research

Table 3 shows the values of the Pearson correlation coefficient describing the level of linear correlation between variables in experiment, in the group of musicians, including years of continuous practice, age when instrument lessons began, results in the great octave, one-line octave, and three-line octave.

Data presented in Table 3 show that there is a statistically negative correlation between the number of years of continuous practice of music and the duration of sound at which sequences fissioned in two perceptual streams in great, one-line, and three-line octaves. There is also a positive statistical correlation between listener age and the above variable (sound duration).

Figure 9 presents scatter plots of the correlations observed in experiment in the group of musicians among the following variables: age, years of continuous practice, age at first instrumental lessons, great octave, one-line octave, and three-line octave.

For a better understanding and elucidation of the obtained experimental data within the musician and non-musician cohorts, as well as to appreciate their significance, it was determined that one more statistical test needs to be performed. A one-way ANOVA analysis of variance, which was opted for in this case, is a statistical method that enables one to examine the observations

Table 3. Values of the Pearson correlation coefficient between variables in the group of musicians, and their statistical significance – experiment

Specification		Years of continuous practice	Age at first instrumental lessons	Great octave	One-line octave	Three-line octave
Years of continuous practice	Pearson correlation coefficient	1	-0.839	-0.599	-0.520	-0.486
	significance (two-tailed)	–	0.000	0.002	0.009	0.016
	N	24	24	24	24	24
Age at first instrumental lessons	Pearson correlation coefficient	-0.839	1	0.727	0.667	0.653
	significance (two-tailed)	0.000	–	0.000	0.000	0.001
	N	24	24	24	24	24
Great octave	Pearson correlation coefficient	-0.599	0.727	1	0.957	0.918
	significance (two-tailed)	0.002	0.000	–	0.000	0.000
	N	24	24	24	24	24
One-line octave	Pearson correlation coefficient	-0.520	0.667	0.957	1	0.981
	significance (two-tailed)	0.009	0.000	0.000	–	0.000
	N	24	24	24	24	24
Three-line octave	Pearson correlation coefficient	-0.486	0.653	0.918	0.981	1
	Significance (two-tailed)	0.016	0.001	0.000	0.000	–
	N	24	24	24	24	24

Source: own research

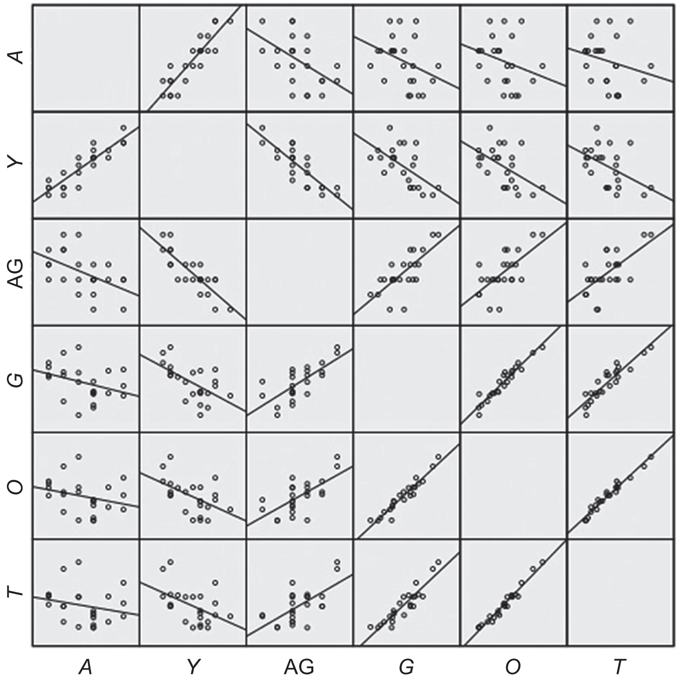

Fig. 9. Scatter plots of correlations observed in experiment in the group of musicians (explanation: *A* – age, *Y* – years of continuous practice, AG – age at first instrumental lessons, *G* – great octave, *O* – one-line octave, *T* – three-line octave)
Source: own research

made in the course of the experiment. With the use of this method, one is able to estimate the probability with which isolated and, at the same time, differentiating factors (results) that actually occur can result in differences between the observed averages in given cohorts. In addition, a univariate analysis examines the influence of each factor in turn, with the latter being considered completely separately during statistical analysis.

Following analysis, the averaged data for both groups of subjects (musicians and non-musicians) compiled in Table 5, and the collated results indicate a high level of statistical significance. This means that the experiment demonstrated tangible and realistically occurring differences in sound perception between the two cohorts for each of the variants. For the great octave, the level of statistical significance was $p < 10^{-5}$, while for the one- and three-line octaves, the level of statistical significance was computed as $p < 10^{-6}$.

Table 4. Results of a one-way analysis of variance ANOVA test performed for the experiment described in this study

Specification		Sum of squares	*df*	Mean square	*F*	Significance level
Experiment						
Great octave	between cohorts	36135.188	1	36135.188	25.699	$p < 10^{-5}$
	within cohorts	64681.125	23	1406.111	–	–
	total	100816.313	24	–	–	–
One-line octave	between cohorts	85598.521	1	85598.521	52.598	$p < 10^{-6}$
	within cohorts	74860.292	23	1627.398	–	–
	total	160458.813	24	–	–	–
Three-line octave	between cohorts	103323.521	1	103323.521	52.477	$p < 10^{-6}$
	within cohorts	90569.958	23	1968.912	–	–
	total	193893.479	24	–	–	–

Source: own research

Discussion of results

The frequency difference between two sounds at which their sequence is perceived as separated in two perceptual streams was named the *trill threshold* by Miller and Heise[14]. Since the sounds played in the present experiment were separated by an interval of a perfect fifth, the accurate name for their relationship, in terms of music theory, is *tremolando threshold*, since trill refers to a quick repetition of sounds separated by a second.

Experiment demonstrated that the tempo of replay at which the perfect fifth tremolando is fissioned by musicians in two perceptual streams is dependent on the octave to which the sounds belong (cf. Fig. 5). The higher the octave (great, one-line, three-line), the shorter the sample duration at which musicians indicated the fission threshold (the faster tempo of the tremolando). The dispersion of the results, expressed by the value of standard deviation, was similar in the group of musicians, independent of the octave in which the sequence was played.

In the group of non-musicians, the perceptual fission occurred at a slower tempo of the tremolando (longer sample duration). In this group, the average

14 G.A. Miller, G.A. Heise, op. cit., pp. 637, 638.

sample duration at the threshold was characterised by a small difference from octave to octave. This means that the pitch of the octave containing the fifth in question in many cases had no bearing on the fission threshold, or did so to a much lesser extent than in the case of musicians. The two groups of listeners also differed in the dispersal of responses, which was much greater in the group of non-musicians than musicians.

It also should be noted that the group of musicians exhibited a statistically significant correlation between the age of musical initiation and the tremolando threshold perceived. This correlation is consistent with reports that cognitive processes concerning pitch perception are especially stimulated during music education in childhood. An example of the influence of childhood music education on perceptual processes is the fact that among musicians possessing absolute pitch, a great majority are individuals who started music education in kindergarten[15].

In the conditions created in the experiment, the deciding factors influencing sound fission in two perceptual streams were the interval between the sounds and the tempo of the sound sequence. In order to achieve fission, one of those parameters must be increased – the interval or the tempo. The values of the tremolando threshold established in experiment can therefore be qualitatively compared to data published by Miller and Heise[16], and by Shonle and Horan[17], while taking into consideration that the above-mentioned authors played their sequences of intervals at a constant tempo, while the variable was the size of the interval, as the listener regulated the frequency of the lower sound. In the work of Miller and Heise, the frequency difference at which the trill threshold occurred was roughly proportional to the frequency of the sequence's lower sound. Thus, the perceptual threshold was found at a similar interval, irrespective of the frequency range in which the sounds were played. Data obtained in the group of non-musicians qualitatively confirm this observation, since they show the tremolando threshold (in the interval of fifth) as unchanging as it is transposed to other octaves.

15 D. Deutsch, T. Henthorn, E. Marvin, H.S. Xu, *Absolute Pitch Among American and Chinese Conservatory Students. Prevalence Differences, and Evidence for a Speech-Related Critical Period*, "Journal of Acoustical Society of America" 2006, vol. 119, Issue 2, pp. 719–722.

16 Ibidem.

17 J.I. Shonle, K.E. Horan, *Trill Threshold Revisited*, "Journal of Acoustical Society of America" 1976, vol. 59, Issue 2, pp. 469–471.

In the work of Shonle and Horan, the trill threshold had a similar value on a linear scale, as the lower sound was changed in a range from 250 to 1000 Hz, which indicated that the threshold interval was diminished in sequences played at higher therefore frequencies. These results are not consistent with the data described by Miller and Heise, nor with the observations made in the present experiment.

The most likely reason for the difference observed between the results of musicians and non-musicians in assessing tremolando threshold is the fact that music education and practice renders the hearing of musicians especially attuned to changes in pitch, and gives them better ability to recognise sounds in fast sequences than non-musicians[18]. As the tempo of the tremolando increases, a musician's perception is able to keep up longer with the changes in sound pitch. For this reason, the tremolando threshold occurs in musicians at a faster tempo than in non-musicians.

Data regarding amateur instrumental music experience on the tremolando threshold may be treated only as preliminary observations. The group of five subjects with amateur music experience exhibited a significant dispersion of results, and the sample size is too small to allow reliable conclusions regarding perceptual fission of the sound stream in this group of listeners.

Conclusion

Routine audiometric tests, conducted while recruiting subjects participating in psychoacoustic tests to check whether the respondents' hearing is otologically normal, are insufficient for this type of research, as they should be extended to include eligibility criteria regarding musical education. The fact that musical education and practice may affect cognitive processes in sound perception should be taken into account when recruiting listeners for psychoacoustic studies, because often identical psychoacoustic experiments repeated on different groups of subjects show that they respond quite differently[19]. In many cases, musical education can be a factor considerably influencing the outcome

18 L.S. Jacobson, L.L. Cuddy, A.R. Kilgour, *Time Tagging. A Key to Musician's Superior Memory*, "Music Perception" 2003, vol. 20, no. 3, pp. 307–313.

19 J.I. Shonle, K.E. Horan, *Trill Threshold Revisited*, "Journal of Acoustical Society of America" 1976, 59, issue 2, pp. 469–471; G.A. Miller, G.A. Heise, *The Thrill Threshold*, "Journal of Acoustical Society of America" 1950, 22, pp. 637, 638.

of an experiment, which should be recorded as important information, as this fact is not taken into account at all when researchers plan psychoacoustic experiments.

The results of psychoacoustic experiments clearly demonstrate that many years of musical education bring substantial benefits, which can be used by musically educated people, unconsciously, throughout their lives in tasks not only strictly related to music. Broadly understood analysis and processing of sounds in the mind, reaching a person from the external environment, indicates that in the case of musicians there is a different type of interpretation of the sounds incoming to the listener. Musicians can group and isolate sounds differently, paying attention to their different characteristics as compared to non-musicians. Thus music education, when practiced universally, can produce very interesting results when used outside of music, during everyday contact with sounds.

In the experiment conducted, the same sound grouping phenomena in terms of quality were observed in musicians and non-musicians. When the sounds were shortened and a certain limit (threshold) of the playback speed was exceeded, the AB–AB sequence was split into two perceptual streams, consisting of a higher (A) and a lower (B) sound.

With the qualitative similarity of the phenomena investigated, significant differences occurred between the groups of musicians and non-musicians as to the threshold value of the rate at which the sequence was split into two streams. In the group of musicians, this threshold occurred at a faster speed of playing the sequence than in the non-musicians.

In most sequences presented to listeners for evaluation in these experiments, the speed of sound reproduction at which the sequence split into two streams was about twice as high in the musician group as in the non-musician group.

The fact that the range of speed at which a sequence of sounds of different pitch is perceived as a single stream is wider in musicians than in non-musicians and includes faster speeds indicates that musicians are able to keep up with faster pitch changes in a sequence than non-musicians. The existence of such a difference is an expected observation, since musicians have extensive analytical pitch listening skills developed as a result of practice in playing instruments and the exercises they participate in ear training classes during their education in music schools and music studies. In addition to the speed at which the sequence was played, the variables in the experiments also included the octave in which the sounds were arranged. Adjustment of these variables significantly affected the results obtained in the group of musicians, while it had no significant effect on the responses provided by non-musicians. Most probably, the ability to perceptually follow the faster pitch changes in the sequence of sounds performed by the tremolando when transposing the interval to a higher octave results from differences in auditory sensitivity to pitch changes within the range of the musical scale. In the examined pitch range, from the great octave to the three-line octave, the hearing sensitivity to pitch changes increases along with the pitch[20]. The effect of the octave in which the sounds are played is presumably revealed when the sequence is played quickly and it is more difficult for the listener to perceptually follow pitch changes in a single stream. In the non-musician group, when the sequences were played at a slower speed, differences in auditory sensitivity to changes in pitch had no effect on the tremolando perception threshold.

It should be emphasised that the differences between the results obtained in the group of musicians and non-musicians were in most cases significant and showed a high level of statistical significance, which is reflected in the analyses presented in the description of the results in the individual chapters, as well as in the summary presented in the appendix.

The fact that individual results in the groups of musicians and non-musicians were significantly spread in all experiments is very important in the interpretation of the results obtained, with the spread being greater in the non-musicians than in the musicians. In analysing the results of the experiments, it is important to remember that the research referred to phenomena related to cognitive processes, which greatly depend on a given person's experience in sound perception, as well as on many other personal characteristics. Reports published by various authors reveal that the grouping of sounds into streams is determined, among other things, by the functional symmetry of the motor and sensory organs and the lateral dominance (laterality) of these organs[21]. These features occur in the population in different configurations.

20 A. Miśkiewicz, A. Rakowski, P. Rogoziński, M. Kocańda, *Progi różnicowe częstości a progi różnicowe wysokości dźwięków muzycznych*, in: A. Rakowski (ed.), Kształtowanie i percepcja sekwencji dźwięków muzycznych, Wydawnictwo Akademii Muzycznej im. Fryderyka Chopina, Warszawa 2001, pp. 125–154; C.C. Wier, W. Jesteadt, D.M. Green, *Frequency Discrimination as a Function of Frequency and Sensation Level*, "Journal of the Acoustical Society of America" 1977, vol. 61, issue 1, pp. 179–184.

21 D. Deutsch, *Two Channel Listening to Musical Scales*, "Journal of the Acoustical Society of America" 1975, vol. 57, issue 5, pp. 1156, 1157; idem, *Ear Dominance and Sequential Interactions*, "Journal of the Acoustical Society of America" 1980, vol. 67, issue 1, pp. 225–227; idem, *The Octave Illusion in Relation to Handedness and Familial Handedness Background*, "Neuropsychologia" 1983, vol. 21, issue 3, pp. 290–292; idem,

The complexity of the phenomena affecting the grouping of sounds into perceptual streams makes it difficult to obtain data that could be normative for a specific type of group of listeners. Therefore, when analysing the results of the experiments, it is important to bear in mind that the results for individuals may differ significantly from the data averaged over a group of listeners.

The conducted research also yields a conclusion relating to the recruitment of listeners participating in experiments on psychoacoustic aspects. If the experiments do not address hearing pathology, the eligibility criteria for listeners in the experimental group usually include the fact that the person has otologically normal hearing, which is checked by routine audiometric testing, and, in some cases, the age of the listener. Contemporary psychoacoustic research is paying increasing attention to issues relating to cognitive processes in sound perception. The results obtained in this study show that in this type of research, it may be very important to consider whether the listeners participating in the research have a musical background.

The experiments conducted in this study can be classified as basic research and were concerned with the perception of elementary sound sequences which may be components of larger sound structures in musical works. The results may also be of practical importance for musicians, as they broaden their knowledge of the principles of sound perception occurring in situations relevant to technical issues faced by composers of electroacoustic music in their work on the realization of works and by sound directors. At the same time, it should be noted that the experiments described in this paper concerned elementary perceptual phenomena occurring during the reproduction of artificially generated test sound sequences to listeners. In musical works, the perception

Auditory Illusions, Handedness, and the Spatial Environment, "Journal of the Audio Engineering Society" 1983, vol. 31, pp. 610–612; idem, *Reply to Reconsidering Evidence for the Suppression Model of the Octave Illusion*, by C.D. Chambers, J.B. Mattingley, S.A. Moss, "Psychonomic Bulletin & Review" 2004, vol. 11, issue 4, pp. 671, 672; idem, *An Auditory Illusion*, "Journal of the Acoustical Society of America" 1974, vol. 55, S18–S19; idem, *The Octave Illusion Revisited Again*, "Journal of Experimental Psychology. Human Perception and Performance" 2004, vol. 30, pp. 359–363; idem, *Illusions for Stereo Headphones*, "Audio Magazine" 1987, March, pp. 43–45; idem, *The Octave Illusion and The What-Where Connection*, in: R.S. Nickerson, R.W. Pew (ed.), *Attention and Performance*, Hillsdale, NJ, Erlbaum 1980, pp. 578–580; D. Deutsch, *Musical Illusions*, "Scientific American" 1975, vol. 233, no. 4, pp. 93–95; D. Deutsch, *An Auditory Illusion*, "Nature" 1974, 251, pp. 307–309.

of sounds as being grouped into streams, for example, as individual voices in a polyphonic work, is the effect of the coexistence of a number of complex processes with underlying interdisciplinary issues, as described in research papers.

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Abstract

This paper discusses an experiment conducted with two groups of participants, composed of musicians and non-musicians, in order to investigate the impact that the speed of a sound sequence and the pitch at which selected sounds are played on the grouping of sounds into perceptual streams. Significant differences were observed between musicians and non-musicians with respect to the threshold sequence speed at which the sequence was split into two streams. In modern psychoacoustic studies, the qualifying criteria for listeners usually include otologically normal hearing (verified by audiometric test) and age. The differences in the results for the two groups suggest that the musical background of the participating listeners may be a vital factor. The criterion of musical education should be taken into account during experiments so that the results obtained are reliable, uniform and free from interpretive errors.

Key words: hearing, psychoacoustics, musical acoustics, music education.